

ANATOMY, PHYSIOLOGY & PATHOLOGY



SYNOPSIS

INTRODUCTION

ORGAN SYSTEM

FLOWCHART & PICTURE

PATHOLOGICAL CONDITION

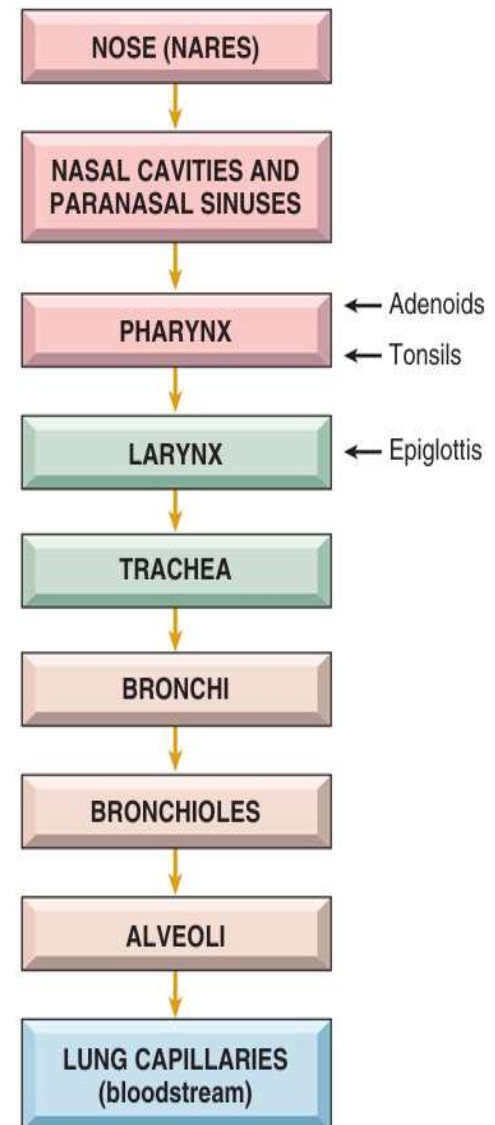
TERMINOLOGY

ICD & CPT

INTRODUCTION

- Anatomy is the study of the structure and form of organisms.
- Physiology is the study of functions and mechanisms of a living being.
- The words, or terms, which make up the language of medicine are referred to as the terminology of the medical field, or medical terminology.

PATHWAY OF RESPIRATORY SYSTEM



RESPIRATORY TERMINOLOGY

MEDICAL TERMINOLOGIES

COMBINING FORMS	MEANING
adenoid/o	adenoids
alveol/o	alveolus, air sac
bronch/o bronchi/o	bronchial tube, bronchus
bronchiol/o	bronchiole, small bronchus

capn/o	carbon dioxide
coni/o	dust
cyan/o	blue
epiglott/o	epiglottis
laryng/o	larynx, voice box
lob/o	lobe of the lung
mediastin/o	mediastinum
nas/o	nose
orth/o	straight, upright
ox/o	oxygen
pector/o	chest
pharyng/o	pharynx, throat
phon/o	voice
phren/o	diaphragm
pleur/o	pleura
pneum/o, pneumon/o	air, lung
pulmon/o	lung
rhin/o	nose
sinus/o	sinus, cavity
spir/o	breathing
tel/o	complete
thorac/o	chest
tonsill/o	tonsils
trache/o	trachea, windpipe

PATHOLOGICAL CONDITION

- Auscultation - Listening to sounds within the body.
- Percussion - Tapping on a surface to determine the difference in the density of the underlying structure.
- Pleural rub - Scratchy sound produced by pleural surfaces rubbing against each other.
- Rales (crackles) - Fine crackling sounds heard on auscultation (during inhalation) when there is fluid in the alveoli.
- Rhonchi (singular: rhonchus) - Loud rumbling sounds heard on auscultation of bronchi obstructed by sputum.
- Sputum - Material expelled from the bronchi, lungs, or upper respiratory tract by spitting.
- Stridor - Strained, high-pitched sound heard on inspiration caused by obstruction in the pharynx or larynx.
- Wheezes - Continuous high-pitched whistling sounds produced during breathing

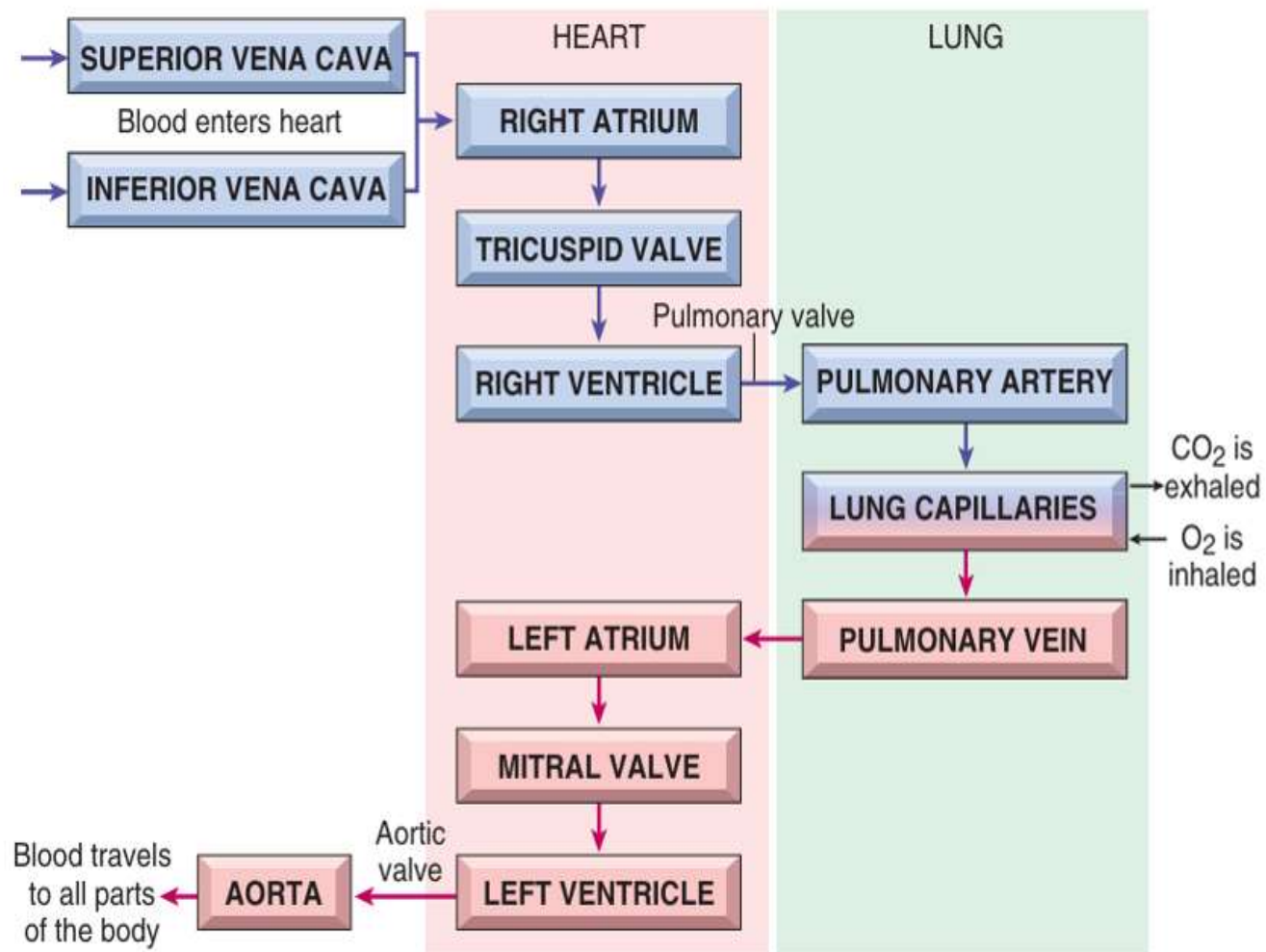
ICD 10 CM

1. Croup
2. Epistaxis
3. Diphtheria
4. Pertussis (Whooping Cough)
5. Asthma
6. Bronchiectasis
7. Chronic Bronchitis
8. Cystic Fibrosis (CF)
9. Atelectasis
10. Emphysema
11. Lung Cancer
12. Pneumoconiosis(Coal worker's)
13. Pulmonary Embolism (PE)
14. Pulmonary Fibrosis
15. Sarcoidosis
16. Tuberculosis (TB)

ANSWER KEY FOR ICD 10 CM

1. J05.0 – Acute obstructive laryngitis (croup)
2. R04.0 -epistaxis
3. A36.9 – Diphtheria, unspecified
4. A37.9 – Whooping cough, unspecified organism
5. J45.909 – Unspecified asthma, uncomplicated
6. J47.9 – Bronchiectasis, uncomplicated
7. J42 – Unspecified chronic bronchitis
8. E84.9 – Cystic fibrosis, unspecified
9. J98.11 – Atelectasis
10. J43.9 – Emphysema, unspecified
11. C34.90 – Malignant neoplasm of unspecified part of unspecified bronchus or lung
12. J60 –pneumoconiosis
13. I26.99 – Other pulmonary embolism without acute cor pulmonale
14. J84.10 – Pulmonary fibrosis, unspecified
15. D86.0 – Sarcoidosis of lung
16. A16.9 – Respiratory tuberculosis, unspecified, without confirmation

PATHWAY OF CARDIOVASCULAR SYSTEM



CONDUCTION SYSTEM OF HEART

SINOATRIAL (SA) NODE /PACEMAKER – UPPER PART OF RT ATRIUM – ATRIA TO CONTRACT



ATRIOVENTRICULAR NODE (AV NODE) IN INTERATRIAL SEPTUM



AV BUNDLE / BUNDLE OF HIS



RT AND LEFT BRANCHES



PURKINJIE FIBRES (VENTRICULAR WALLS CONTRACT)



FORCES BLOOD OUT OF AORTA AND PULMONARY ARTERY

PATHOLOGICAL CONDITIONS

- CONGESTIVE HEART FAILURE (CHF) - Heart is unable to pump its required amount of blood.
- Coronary artery disease (CAD) - Disease of the arteries surrounding the heart – MAINLY OF ATHEROSCLEROSIS
- Endocarditis - Inflammation of the inner lining of the heart.
- Hypertensive heart disease - High blood pressure affecting the heart.
- Cardiomyopathy - Any disease of the heart muscle.
- Cardiac arrest (CA) - Failure of the systemic circulatory system due to absent or inadequate contraction of the ventricles
- COR PULMONALE - Right ventricular enlargement that is caused by a lung disorder. It leads to hypertension in the pulmonary artery.
- Mitral Valve Prolapse (MVP) Flaps of mitral valve extend back into L atrium causes leaking Mostly genetic basis

MEDICAL TERMINOLOGIES:

COMBINING FORM	MEANING
angi/o	vessel
aort/o	aorta
arter/o , arteri/o	artery
atri/o	atrium
ather/o	yellowish plaque
brachi/o	arm
cardi/o	heart
coron/o	heart
isch/o	hold back
phleb/o	vein
sphygm/o	pulse
thromb/o	clot
valv/o	valve
vas/o	vessel; duct; vas deferens
vascul/o	blood vessel

ICD 10 CM

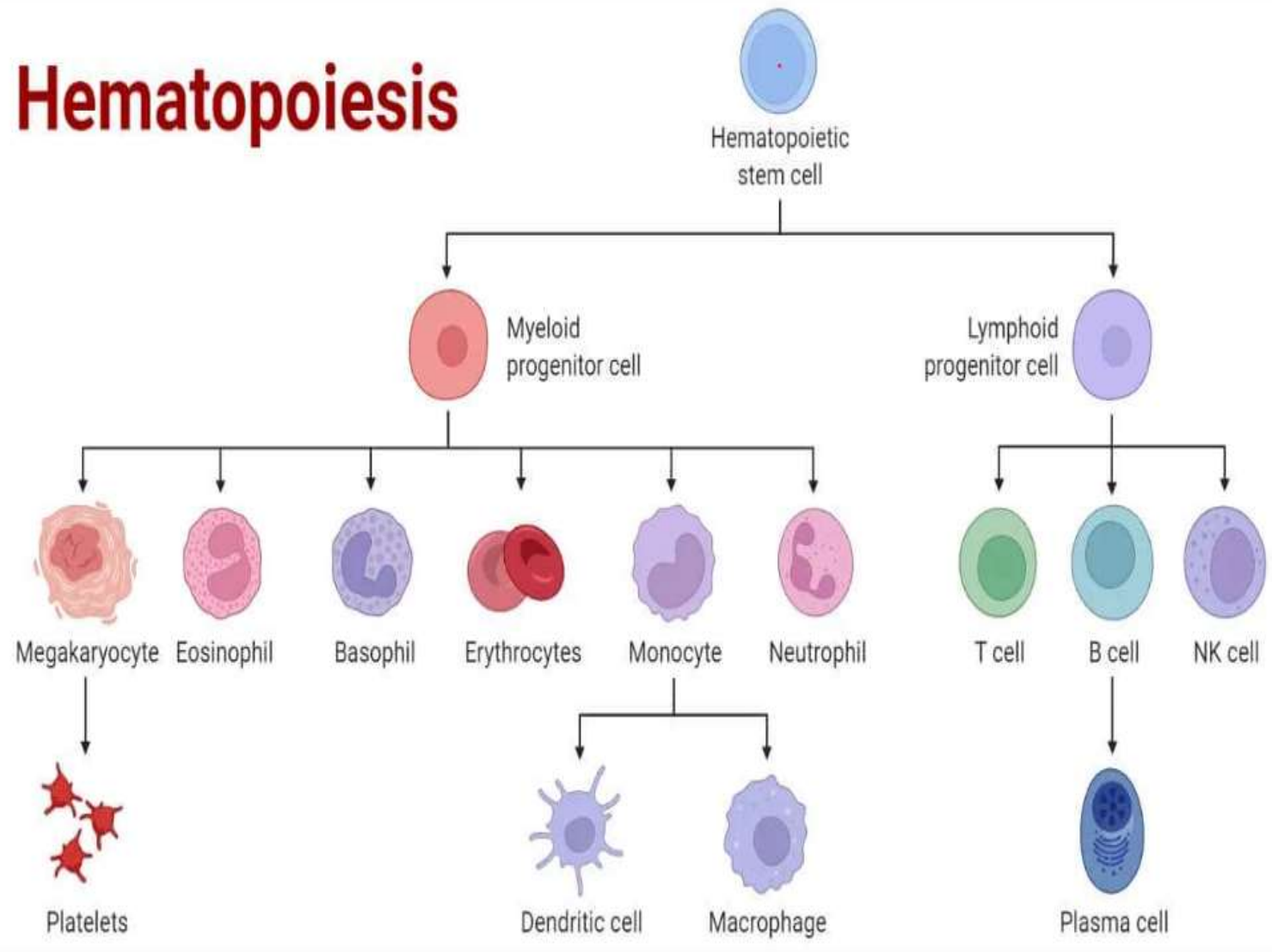
- Aneurysm
- Angina pectoris
- Other causes include coronary embolism.
- Arteriovenous fistula
- Arteritis Inflammation of one or several arteries
- Atherosclerosis
- Atrial septal defect (ASD)
- Bruit
- Bundle branch block (BBB)
- Cardiac arrest (CA)
- Cardiomyopathy
- Congestive heart failure (CHF)
- Myocardial infarction.
- Coronary artery disease (CAD)
- Embolism

ANSWER KEY FOR ICD 10 CM

- Aneurysm-I72.9 – Aneurysm, unspecified
- Angina Pectoris-I20.9 – Angina pectoris, unspecified
- Coronary Embolism (Other causes of cardiac ischemia) I24.19 – Other forms of acute ischemic heart disease
- Arteriovenous Fistula-I77.0 – Arteriovenous fistula, acquired,
- Arteritis (Inflammation of arteries)-I77.6 – Arteritis, unspecified
- Atherosclerosis-I70.90 – Atherosclerosis, unspecified
- Atrial Septal Defect (ASD)-Q21.1 – Atrial septal defect
- Bruit-R09.89 – Other specified symptoms and signs involving the circulatory and respiratory systems
- Bundle Branch Block (BBB)-I45.19 – Other right bundle-branch block
- Cardiac Arrest (CA)- I46.9 – Cardiac arrest, cause unspecified
- Cardiomyopathy-I42.9 – Cardiomyopathy, unspecified
- Congestive Heart Failure (CHF)-I50.9 – Heart failure, unspecified
- Myocardial Infarction (MI)-I21.9 – Acute myocardial infarction, unspecified
- Coronary Artery Disease (CAD)-I25.119 – With unspecified angina pectoris
- Embolism-I74.9 – Arterial embolism and thrombosis, unspecified

PATHWAY OF BLOOD SYSTEM

Hematopoiesis



PATHOLOGICAL CONDITION

- Any abnormal or pathologic condition of the blood - blood dyscrasia (disease).
- Anemia - Deficiency in erythrocytes or hemoglobin.
- Aplastic anemia - Failure of blood cell production in the bone marrow – Pancytopenia
- Hemolytic anemia - Reduction in red cells due to excessive destruction – Eg : congenital spherocytic anemia (hereditary spherocytosis)
- Pernicious anemia - Lack of mature erythrocytes caused by inability to absorb vitamin B12 into the bloodstream.
- Sickle cell anemia - Hereditary disorder of abnormal hemoglobin producing sickle shaped erythrocytes and hemolysis.
- Thalassemia - Inherited defect in ability to produce hemoglobin, leading to hypochromia. Thalassa is a Greek word meaning sea.
- Hemochromatosis - Excess iron deposits throughout the body.
- Polycythemia vera - General increase in red blood cells (erythremia).

MEDICAL TERMINOLOGIES

COMBINING FORM	MEANING
bas/o	base (<i>alkaline</i> ,
chrom/o	color
coagul/o	clotting
cyt/o	cell
eosin/o	red, dawn, rosy
erythr/o	red
granul/o	granules
hem/o	blood
hemoglobin/o	hemoglobin
is/o	same, equa
kary/o	nucleus
leuk/o	white
mon/o	one, single

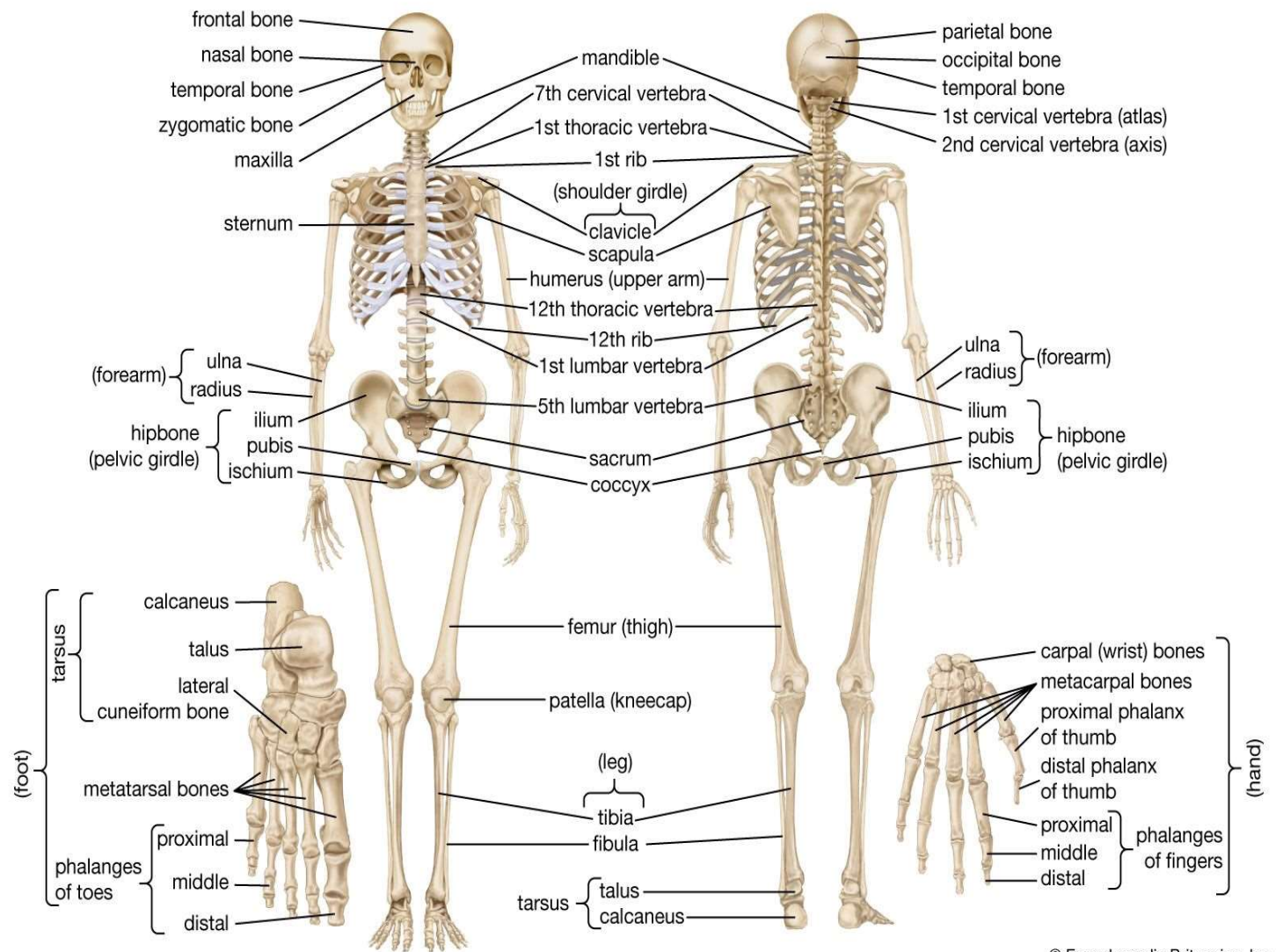
ICD 10 CM

1. Hemophilia A
2. Von Willebrand Disease
3. Thrombocytopenia
4. Thrombocytosis
5. Hemochromatosis
6. Polycythemia Vera
7. Disseminated Intravascular Coagulation (DIC)
8. Leukemia – Acute Lymphoblastic (ALL)
9. Myelodysplastic Syndromes (MDS)
10. Lymphoma – Hodgkin's Disease
11. Multiple Myeloma
12. Pancytopenia
13. Neutropenia
14. Eosinophilia
15. Hemolytic Uremic Syndrome (HUS)

ANSWER KEY

1. Hemophilia A -D66 – Hereditary factor VIII deficiency
2. Von Willebrand Disease -D68.0 – Von Willebrand's disease
3. Thrombocytopenia-D69.6 – Thrombocytopenia, unspecified
4. Thrombocytosis- D75.1 – Essential thrombocythemia
5. Hemochromatosis-E83.11 – Hemochromatosis (iron overload disorder)
6. Polycythemia Vera-D45 – Polycythemia vera (a myeloproliferative disorder)
7. Disseminated Intravascular Coagulation (DIC)-D65 – Disseminated intravascular coagulation
8. Leukemia – Acute Lymphoblastic (ALL)-C91.0x – Acute lymphoblastic leukemia (x denotes remission status)
9. Myelodysplastic Syndromes (MDS)-D46.9 – Myelodysplastic syndrome, unspecified
10. Lymphoma – Hodgkin’s Disease-C81.90 – Hodgkin lymphoma, unspecified, unspecified site
11. Multiple Myeloma-C90.00 – Multiple myeloma not having achieved remission
12. Pancytopenia-D61.818 – Other pancytopenia (reduction in RBCs, WBCs, and platelets)
13. Neutropenia-D70.9 – Neutropenia, unspecified
14. Eosinophilia-D72.1 – Eosinophilia
15. Hemolytic Uremic Syndrome (HUS)-D59.3 – Hemolytic-uremic syndrome

MUSCULOSKELETAL SYSTEM



PATHOLOGICAL CONDITIONS

Pathology

Condition or Disease	Description
Achilles tendonitis	Inflammation of the achilles tendon.
Ankylosing Spondylitis	Chronic systemic arthritis involving inflammation and stiffening of the axial skeleton and the large peripheral joints.
Ankylosis	Stiffening or fixation of a joint.
Arthralgia	Pain in a joint. Also called <i>arthrodynia</i> .
Arthritis	A variety of conditions involving inflammation of the joints.
Arthropathy	Any disease that affects a joint.
Articular disease	Any disease involving the joints. These conditions include arthritis, Rheumatoid arthritis, osteoarthritis, gout, and bursitis.
Avascular necrosis	Death of bone tissue caused by inadequate blood supply. Can be caused by several conditions such as trauma, fracture, osteoarthritis, and systemic lupus erythematosus. Can be idiopathic. Also called <i>osteonecrosis</i> .

Bunion	A type of bursitis that affects the joint at the base of the great toe.
Bursitis	An acute or chronic inflammation of the bursa. Occurs most commonly in the shoulders and elbows, but can occur in any bursa.
Carpal tunnel Syndrome (cts)	Compression of the median nerve as it passes through the tunnel in the wrist containing the median nerve and blood vessels. It is frequently caused by repeated flexion and extension of the wrist. Can lead to pain, tingling, and numbness
Claudication Limping.	Usually intermittent.
Discoid lupus Erythematosus (dle)	A chronic disorder of unknown etiology that primarily affects the skin. Qcharacterized by macules, plaques, and scales that appear on the skin.
Ewing tumor	A malignant bone tumor that occurs most commonly in individuals between 10 and 20 years of age. Occurs most often in long bones in the extremities. Tends to involve extensive portions of a bone. Also called ewing sarcoma.

MEDICAL TERMINOLOGY

Medical terminologies:

Combining forms	Meaning
calc/o, calci/o	calcium
kyph/o	humpback, hunchback (posterior curvature in the thoracic region)
lamin/o	lamina (part of the vertebral arch)
lord/o	curve, swayback (anterior curvature in the lumbar region)
lumb/o	loins, lower back
myel/o	bone marrow
orth/o	straight
oste/o	bone
scoli/o	crooked, bent (lateral curvature)
spondyl/o (used to make words about conditions of the structure)	vertebra
vertebr/o (used to describe the structure itself)	vertebra

ICD 10 CM

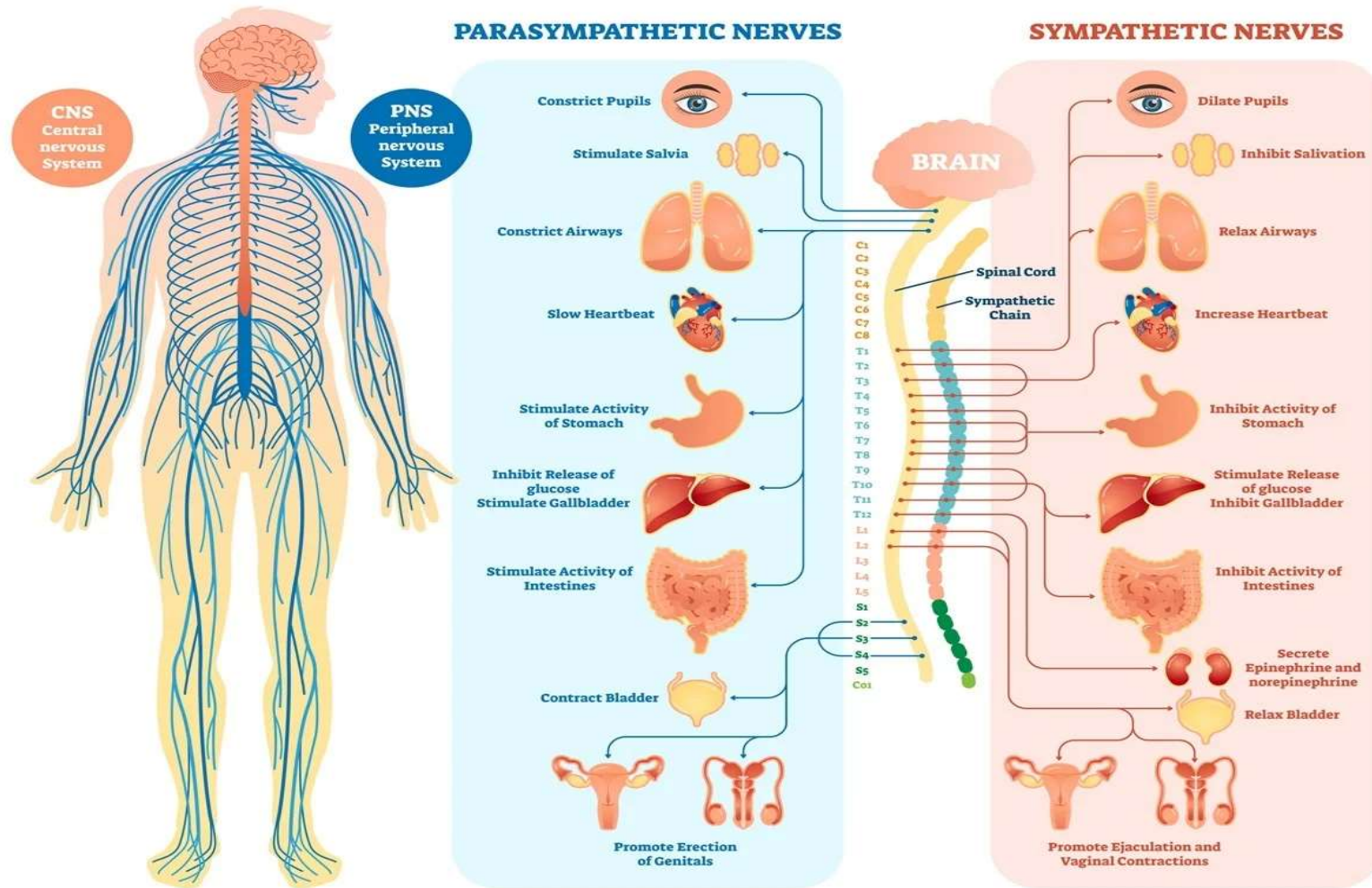
- Osteoarthritis
- Rheumatoid Arthritis (RA)
- Osteoporosis (without fracture)
- Gout-
- Lupus (Systemic Lupus Erythematosus with organ involvement)-
- Fibromyalgia-
- Ankylosing Spondylitis
- Bursitis
- Tendinitis
- Rotator Cuff Tear
- Meniscus Tear (Knee)
- Carpal Tunnel Syndrome
- Sciatica-
- Herniated Disc (Intervertebral disc displacement)-
- Scoliosis-
- Kyphosis-
- Lordosis-
- Muscle Strain (e.g., lower back)-
- Fracture of Femur (unspecified)-
- Myositis (inflammation of muscle)-

ANSWER KEY

- Osteoarthritis (OA)-M19.90 – Unspecified osteoarthritis, unspecified site
- Rheumatoid Arthritis (RA)-M06.9 – Rheumatoid arthritis, unspecified
- Osteoporosis (without fracture)-M81.0 – Age-related osteoporosis without current pathological fracture
- Gout-M10.9 – Gout, unspecified
- Lupus (Systemic Lupus Erythematosus with organ involvement)-M32.10 – Systemic lupus erythematosus, organ or system involvement unspecified
- Fibromyalgia-M79.7 – Fibromyalgia
- Ankylosing Spondylitis-M45.9 – Ankylosing spondylitis of unspecified sites
- Bursitis-M71.9 – Bursitis, unspecified
- Tendinitis-M77.9 – Enthesopathy, unspecified (commonly used for tendinitis)
- Rotator Cuff Tear-M75.121 – Complete rotator cuff tear or rupture of right shoulder
- Meniscus Tear (Knee)-S83.241A – Complex tear of medial meniscus, right knee, initial encounter
- Carpal Tunnel Syndrome-G56.01 – Carpal tunnel syndrome, right upper limb
- Sciatica-M54.30 – Sciatica, unspecified side
- Herniated Disc (Intervertebral disc displacement)-M51.26 – Other intervertebral disc displacement, lumbar region
- Scoliosis-M41.9 – Scoliosis, unspecified
- Kyphosis-M40.209 – Unspecified kyphosis, site unspecified
- Lordosis-M40.5 – Other lordosis
- Muscle Strain (e.g., lower back)-S39.012A – Strain of muscle, fascia and tendon of lower back, initial encounter
- Fracture of Femur (unspecified)-S72.90XA – Unspecified fracture of femur, initial encounter
- Myositis (inflammation of muscle)-M60.9 – Myositis, unspecified

NERVOUS SYSTEM

HUMAN NERVOUS SYSTEM



PATHOLOGICAL CONDITIONS

Condition or Disease	Description
absence seizure	A 10- to 30-second loss of consciousness, typically marked by eyelid fluttering. The patient abruptly stops activity but does not fall. After the seizure, the patient resumes activity without knowledge that the seizure even occurred. Formally known as <i>petit mal seizures</i> .
agraphia	The inability to write properly in the absence of any physical problems associated with the limbs.
Alzheimer disease	A progressive loss of cognitive function, memory, and ability to calculate. The patient becomes confused and disoriented. Associated with loss of neurons and an excessive number of axonal plaques in the cerebral cortex and subcortical structures.
amnesia	Disturbance in long-term memory, characterized by a partial or total inability to recall past experiences.

amyotrophic lateral sclerosis (ALS)	A progressive disorder resulting in destruction of motor neurons. Also called <i>Lou Gehrig disease</i> .
aphasia	The inability to comprehend or express words due to injury or degeneration of language centers in the cerebral cortex. May affect speech, writing, or sign communication.
apraxia	The inability to perform skilled or purposeful motor acts that were previously learned, despite the willingness and physical ability to perform them. Due to cerebral disorders.
ataxia	The inability to control muscle activity during voluntary movement. Movements become jerky. Caused by disorders affecting the cerebellum or spinal cord.
Bell palsy	Sudden onset of facial paralysis that is typically unilateral and without any known cause. It is believed to be caused by immune disorder or viral disease that affects cranial nerve VII.

MEDICAL TERMINOLOGY

MEDICAL TERMINOLOGIES:

Form	Meaning
cerebr/o	cerebrum
crani/o	cranium (skull)
encephal/o	brain
gangli/o	nervous
mening/o	membranes; meninges
myel/o	spinal cord; bone marrow
neur/o	nerve

ICD 10 CM

- Agraphia
- Alzheimer Disease
- Amnesia
- Amyotrophic Lateral Sclerosis (ALS)
- Aphasia
- Apraxia
- Ataxia
- Bell Palsy

Condition	ICD-10-CM Code(s)
Agraphia	R48.2 – Apraxia (Note: Agraphia is often classified under apraxia)
Alzheimer Disease	G30.9 – Alzheimer's disease, unspecified
Amnesia	R41.3 – Other amnesia
Amyotrophic Lateral Sclerosis (ALS)	G12.21 – Amyotrophic lateral sclerosis
Aphasia	I69.320 – Aphasia following cerebral infarction
Apraxia	R48.2 – Apraxia
Ataxia	G11.9 – Hereditary ataxia, unspecified
Bell Palsy	G51.0 – Bell palsy

Thank
you